



Connecting products to the IoT



Thingsquare Mist is a software system that helps add connectivity to products, enabling them to connect to the IoT. It builds on well-known Open Source components, has a small memory footprint and low computational requirements, and works with multiple microcontrollers with a range of radios. Thingsquare Mist uses an innovative 'sleepy mesh' technology that is extremely efficient and helps increase reliability and range in wireless sensor networks without sacrificing battery life.

Recording sensor data to the Cloud



Nimbits is a collection of software for recording time series data to the Cloud. It helps send sensor information to the Cloud, and connect everything to everything. It powers what they call an Open Source IoT on a distributed Cloud. When systems send new numeric, JSON or XML data into a Nimbits data point using REST Web services, the new values trigger cascading calculations, alerts, statistics and more. The Nimbits server can be used as the backend to applications, generating charts and data visualisations with JavaScript, using the public cloud or your own infrastructure.

"As the Internet of Things evolves further, the apps revolution will pervade into consumer goods"

Shivesh Vishwanathan, senior consultant (Mobility), Persistent Systems, speaks to Janani Gopalakrishnan Vikram about how the definition of the Internet of Things (IoT) has transformed from being RFID-centric to app-centric. He also discusses the characteristics of IoT software and the challenges in developing these

How has the definition of the IoT changed over time? At one time, people were talking about just RFID. But now it appears as if the IoT is fuelled more by smartphones and their apps, than anything else. What do you think?

In the evolution of the IoT, we have gone from short-range, specific communication of proprietary devices to smartphones and tablets. However, as the field evolves further, the apps revolution will pervade beyond smartphones and tablets and into consumer goods like televisions, refrigerators, home automation systems and others; what we call Consumerware—consumerisation of software.

While the IoT started from a hardware perspective, software-based solutions like apps, app stores, location, Web-based protocols and others will fuel the future growth of Consumerware and the IoT. So the definition of the IoT has expanded from being centred around just short-range, RFID-centric device identification to encompass various categories of consumer devices through which intelligent devices as well as people can interact with each other.

What kind of mobile applications are fuelling the concept of the IoT?

The apps for the IoT are still in nascent stages. There are smart refrig-

erators from players like LG and Samsung; and then there are Google Glass and Pebble Watch that act as the entry point for IoT capabilities. However, the challenge with deploying meaningful systems using IoT on multiple devices is that the various endpoints need to be put together to provide a service. These require complementary server systems, ideally based on Cloud Computing technologies, to be put in place.

Location-based apps are apparently very popular. Apart from that, could you give some other examples of, say, the IoT in healthcare?

In healthcare field, Persistent is working with medical device companies to build smart applications that interact with implants that are capable of monitoring bodily functions and, in some cases, also perform actions like delivering mild shocks to the heart. These devices usually use proprietary communication technologies to interact with mobile devices, tablets and even Web applications.

What are the special challenges related to developing IoT applications?

The support systems, including frameworks, regulations, privacy and security, are important considerations, apart from the server-side enablement that is required and the end-to-end use-cases that need to be developed.

Read more about Internet of Things at internetofthings.electronicshub.com